



UNIVERSITY OF
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Superconductivity

Lecture 14 – “SQUID”

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Previously, Josephson junction

DC Josephson effect:

$$I = I_0 \sin \theta_{21}.$$

- Current flows without a voltage
- Current is driven by the phase difference.

AC Josephson effect:

$$I(t) = I_0 \sin \left(\theta_{21}(0) + \frac{2e}{\hbar} V t \right).$$

- A constant voltage leads to AC currents;
- $f = (2e/h)V$;
- 483 597.848 4 GHz/V (Josephson constant)

Previously, Josephson voltage standard

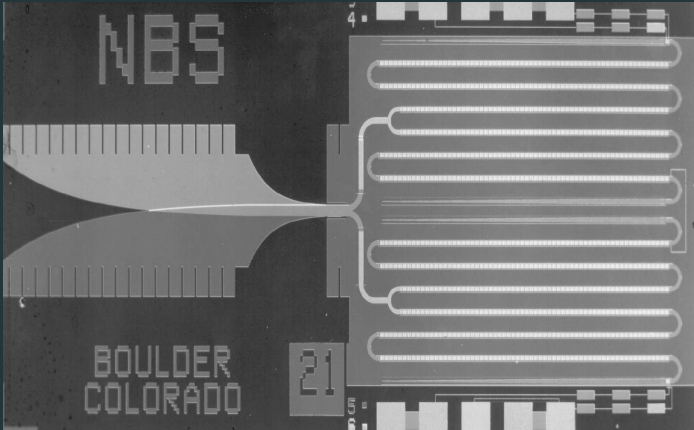


Figure 1: 3020 junctions, NIST

A Josephson junction in a field

Superconducting Quantum Interference Device

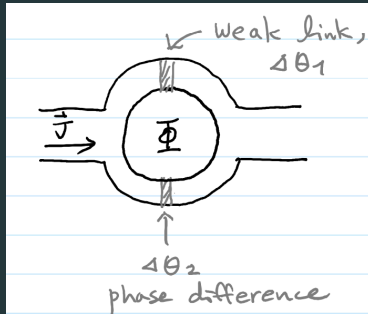


Figure 2: Two Josephson junction on a superconducting ring

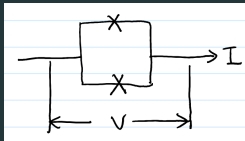
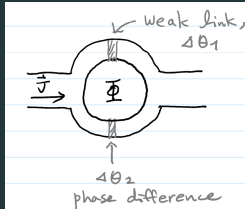


Figure 3: Circuit drawing of a SQUID

Flux through SQUID



Integrate the gauge-invariant phase around a closed path C

$$\oint_C \left[\nabla\theta - \frac{2e}{\hbar} \mathbf{A} \right] \cdot d\mathbf{r} = 2\pi n$$

$$\Delta\theta_1 - \Delta\theta_2 - \frac{2e}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{r} = 2\pi n$$

$$\Delta\theta_1 - \Delta\theta_2 = 2\pi \left(n + \frac{\Phi}{\Phi_0} \right)$$

Current across a SQUID

$$I = I_{c1} \sin(\Delta\theta_1) + I_{c2} \sin(\Delta\theta_2), \quad (1)$$

Assume $I_c = I_{c1} = I_{c2}$.

$$\begin{aligned} I &= I_c \sin(\Delta\theta_1) + I_c \sin(\Delta\theta_2) \\ &= I_c \sin\left(\frac{\Delta\theta_1 + \Delta\theta_2}{2}\right) \cos\left(\frac{\Delta\theta_1 - \Delta\theta_2}{2}\right) \\ &= 2I_c \sin\left(\frac{\Delta\theta_1 + \Delta\theta_2}{2}\right) \cos\pi\left(n + \frac{\Phi}{\Phi_0}\right) \\ &= 2I_c \cos\left(\frac{\pi\Phi}{\Phi_0}\right) \sin(\Delta\theta) \end{aligned}$$

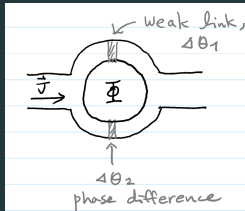
Superconducting Quantum Interference Device

$$I = 2I_c \cos\left(\frac{\pi\Phi}{\Phi_0}\right) \sin(\Delta\theta)$$

$$I_c^{new} = 2I_c \cos\left(\frac{\pi\Phi}{\Phi_0}\right), \quad (2)$$

instead of $2I_c$ for the two junctions.

Superconducting Quantum Interference Device



$$I_c^{new} = 2I_c \cos \left(\frac{\pi\Phi}{\Phi_0} \right), \quad (3)$$

instead of $2I_c$ for the two junctions.

Superconducting Quantum Interference Device

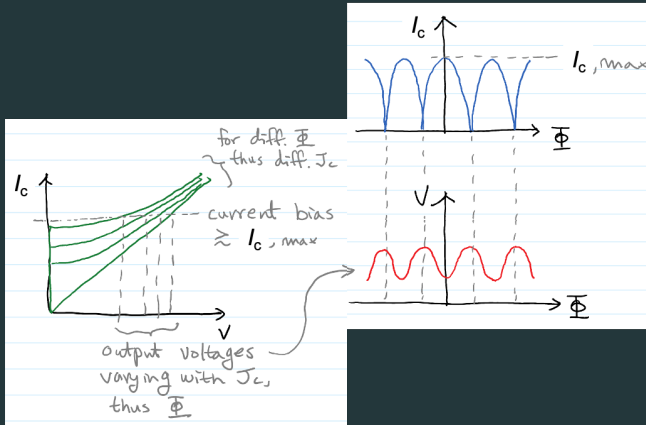


Figure 4: Left: current versus voltage graph. Right upper: critical current as a function of magnetic flux. Right lower: voltage variation as a function of the flux.

SQUID sensitivity

- Earth magnetic field?→ about 5×10^{-5} T.
- Flux quantum $\Phi_0 = h/2e$?→ about 2×10^{-15} Wb.
- 1 mm² ring of SQUID accommodate 'how many' $\Phi_0 \rightarrow 25\,000 \Phi_0$!
- A SQUID magnetometer→ 10^{-14} T.
- Too good to be useful?→ brain activity scale!

MEG (Magneto-Encephalo-Graphy)



MEG scanner



Figure 6: About 300 scanners, Centre for Human Brain Health, University of Birmingham.

A Josephson junction in a field

SQUID critical currents

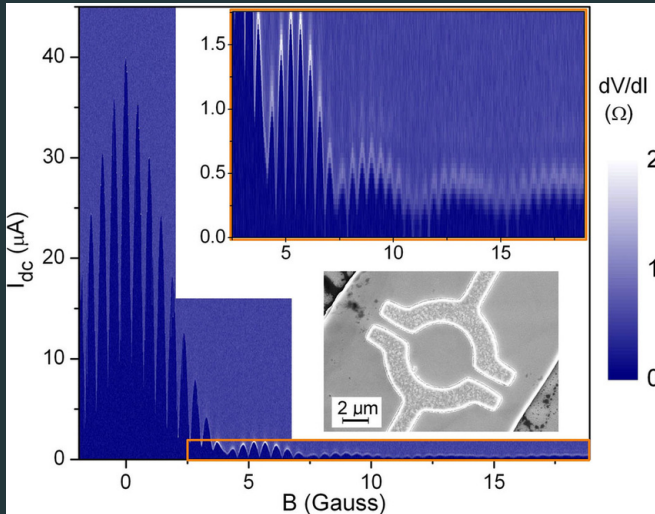
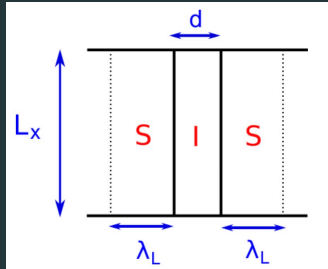


Figure 7: Scientific Reports, 2012

Finite size Josephson junction



$$I_{\max} = I_c \left| \frac{\sin(\pi\Phi'/\Phi_0)}{(\pi\Phi'/\Phi_0)} \right|$$

Fraunhofer pattern

$$I_{\max} = I_c \left| \frac{\sin(\pi\Phi'/\Phi_0)}{(\pi\Phi'/\Phi_0)} \right|$$

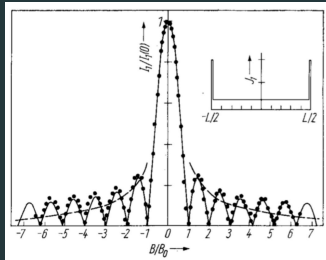


Figure 8: A. Barone et al., Diffraction and interference patterns in single Josephson junctions, Phys. Stat. Sol.(a) 41, 393 (1977).